ECOWATCH: AN INTELLIGENT GEOLOCATION-BASED INFRASTRUCTURE WITH AUTOMATED SOLID WASTE DETECTION AND SPATIAL ANALYSIS FOR THE CITY OF SAN JOSE DEL MONTE

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INTRODUCTION

The problem and its background refer to the context and description of an issue or challenge that requires attention or investigation. It sets the stage for understanding the problem's significance, its underlying causes, and the reasons why it needs to be addressed. Providing a clear background helps to establish the context and enables researchers or problem solvers to develop appropriate strategies and solutions.

Background of the Study

Environmental sustainability is an important concern in the City of San Jose del Monte (CSJDM), one of the fastest-growing residential areas in Bulacan, is environmental sustainability. Because of its topography and fast growth, the city is faced with the problem of illegal dumping of trash in the creeks and vacant lots in the periphery of the city, which hinders the flow of water in the creeks and causes flooding. At present, the reporting of these conditions is done manually, which is time-consuming, as the residents have difficulty in giving the correct location, thereby delaying the response of the City Environment and Natural Resources Office (CENRO).

The researchers want to address this issue by developing an intelligent infrastructure called as "EcoWatch" to help with environmental danger response. By using Instance Segmentation (Mask R-CNN) for automatic visual verification of solid waste, the system goes beyond traditional human reporting. Additionally, the DBSCAN (Density-Based Spatial Clustering) method is used by the system to apply spatial intelligence and convert unprocessed report data into useful "hotspot" clusters. The system uses Ray-Casting (Point-in-Polygon) logic to cross-reference coordinates with digital barangay borders in order to guarantee jurisdictional responsibility. This ensures that each confirmed report is immediately sent to the appropriate local responders.

Project Context

San Jose del Monte continues to expand as a residential and industrial community. With this expansion, environmental issues such as illegal dumping in waterways and abandoned lots remain prevalent in many barangays. Although residents can report problems via hotlines or social media, these channels are frequently manual and inconsistent. Many reports may not include the correct location information, leading the City Environment and Natural Resources Office (CENRO) to encounter delays when confirming and responding to complaints.

This demonstrates a disconnect between SJDM's environmental needs and the lack of a customized, technology-based reporting system tailored to the city's geographic and administrative structure. Current reporting techniques rely primarily on written descriptions and do not include high-accuracy GPS validation or digital boundary checking. The EcoWatch system serves as an intelligent geolocation-based infrastructure designed to bridge this gap.

Purpose and Description

The study was conducted in order to help the City of San Jose del Monte improve environmental monitoring and response by developing the EcoWatch infrastructure. The system has the following capabilities: it provides a mobile-responsive web interface accessible via QR-code-tagged physical markers , it utilizes the W3C Geolocation API for high-precision GPS metadata capture to solve location precision gaps, it performs automated verification of waste in uploaded photos using an Instance Segmentation Model (Mask R-CNN), it employs a Geolocation Routing Engine using Ray-Casting (Point-in-Polygon) logic to automatically dispatch reports based on digital barangay boundaries.

Once the system is deployed in the community, this will give benefits to the following individuals:

SJDM Netizens. They will benefit from this study because it provides a seamless, QR-code-accessible platform that ensures their environmental reports are accurately tagged and routed without the need for complex manual descriptions. The system allows residents to track the status of their reports in real-time, fostering better community engagement and trust in local government responses.

CENRO Administrators. They will surely benefit from this study because the system provides a data-driven dashboard that filters out invalid reports through automated verification. The integration of spatial intelligence allows officials to identify high-density dumping clusters, enabling more effective resource allocation and proactive environmental management for the city.

Researchers. The knowledge that the researchers learn during the development of this study will be beneficial for the future. Mastering advanced IT implementations such as Mask R-CNN, Bi-LSTM, and spatial clustering algorithms serves as a significant asset for their professional careers in the technology industry.

Future Researchers. This study serves as a guide or foundation for innovative technology in municipal environmental management. Future researchers can utilize the framework of EcoWatch to explore further enhancements in automated hazard detection or adapt the geolocation routing logic to solve other administrative challenges within local government units.

Statement of the Problem

The City of San Jose del Monte (SJDM) lacks a centralized, technology-driven infrastructure to manage illegal dumping incidents, leading to a disconnect between citizen reports and government action. To ensure the effectiveness of the EcoWatch system, the

study aims to answer the following questions:

1. How can a localized reporting entry-point be developed using QR-code-tagged markers and high-precision GPS metadata capture?
2. How do the researchers integrate an Instance Segmentation Model (Mask R-CNN) to automatically verify and filter solid waste reports?
3. How do the researchers utilize Ray-Casting (Point-in-Polygon) logic to cross-reference coordinates with digital barangay boundaries for automated dispatching?
4. How can the system employ DBSCAN (Density-Based Spatial Clustering) to transform individual reports into actionable spatial intelligence for city administrators?

Objectives of the Study

The study aims to provide an intelligent geolocation-based infrastructure for automated waste detection and jurisdictional routing in SJDM. The general goal of this study is to design, develop, and implement the EcoWatch system through the following specific objectives:

1. To develop a localized reporting entry-point for SJDM netizens through the deployment of QR-code-tagged physical markers, utilizing the W3C Geolocation API for high-precision GPS metadata capture.
2. To integrate an Instance Segmentation Model (Mask R-CNN) to automatically verify the presence of waste in uploaded photos, filtering out invalid reports to ensure data integrity for city officials.
3. To utilize a Geolocation Routing Engine using Ray-Casting (Point-in-Polygon) logic to cross-reference coordinates with digital barangay boundaries, ensuring reports are automatically dispatched to the correct local responders.

4. To create an Incident Dashboard for SJDM administrators that uses DBSCAN (Density-Based Spatial Clustering) to pinpoint high-density dumping clusters, transforming individual reports into actionable spatial intelligence.

Scope and Limitation

The following are some of the limitations of the proposed system: it is specifically tailored for the administrative and geographic layout of the City of San Jose del Monte (CSJDM). The main reason is that the jurisdictional routing and spatial analysis algorithms require precise digital barangay boundaries and localized GPS metadata to function effectively. The only way to submit an environmental report is through the use of QR-code-tagged physical markers, which serve as the localized entry point for the mobile-responsive interface. The system can verify and detect solid waste only due to the technical constraints of the training dataset for the instance segmentation model; for example, it cannot currently detect liquid chemical spills or air quality hazards. The system is electricity and internet dependent for its AI-driven features; any signs of disruption may result in the Mask R-CNN verification engine being temporarily unavailable. The system is still usable for capturing basic GPS coordinates if the residents do not require immediate automated verification or conversational updates.

The precision of the automated dispatch is not guaranteed to be absolute if the digital boundaries provided by the LGU are overlapping or inaccurate. If the user's mobile device has a hardware-limited GPS sensor, there may be slight discrepancies in the captured coordinates. The system also can only measure and analyze data points related to waste detection and spatial clustering; these are the only high-level machine learning implementations currently utilized by the researchers.